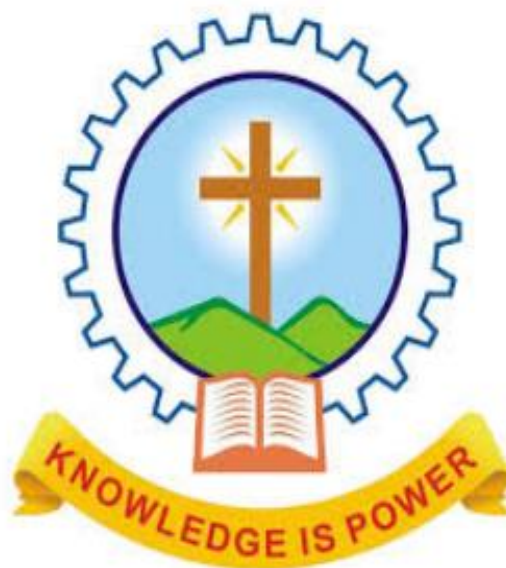

**MAR ATHANASIUS
COLLEGE OF ENGINEERING
(AUTONOMOUS)
KOTHAMANGALAM**



.....

PRACTICAL RECORD

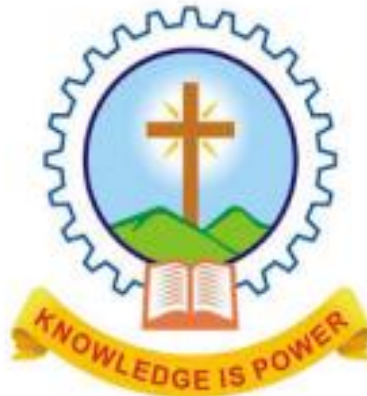
Year:

NAME: JEWEL THOMAS

CLASS No.: 42 BATCH: S2 AH

REG. No.: OF:

**MAR ATHANASIUS
COLLEGE OF ENGINEERING
KOTHAMANGALAM**



PRACTICAL RECORD

Year:

NAME: A JEWEL THOMAS
CLASS No: 1C 42 BATCH: S2 AH

Uni. Ex: Reg No. of March/Sept.

Uni. Ex: Reg No. of March/Sept.

*Certified that this is the Bonafide Record of Practical work done in the Laboratory of
M. A. College of Engineering Kothamangalam
by.....*

Kothamangalam,

Staff Member-in-Charge

.....

External Examiner

Internal Examiner

INDEX

[illegible]

EXPERIMENT – 1

Aim- Familiarise the installation procedure and document preparation in LaTeX

Theory -LaTeX, is a document preparation system for high-quality typesetting. It is most often used for medium-to-large technical or scientific documents but it can be used for almost any form of publishing. LaTeX is not a word processor! Instead, LaTeX encourages authors not to worry too much about the appearance of their documents but to concentrate on getting the right content

Procedure-

5. Document Preparation

All LaTeX documents have a standard structure

```
\documentclass[11pt,a4paper]{article}
\usepackage[T1]{fontenc}
\begin{document}

\end{document}
```

% The part between

\documentclass

preamble

\begin{document}

% The part between

\begin{document}

body

\end{document}

While processing the input file, LaTeX needs to know what size of the paper the author intends to create.

`\documentclass[options](article).`

We can provide the features required for our document in the options section.

Class indicates which document is intended to be created.

IMPLEMENTATION-

```
\documentclass{article}

\usepackage{graphicx} % Include graphicx package for image handling

\begin{document}

% Suppress the header and page number for the title page

\thispagestyle{empty}

\title{\textbf{My First LaTeX Document} \\ \large JEWEL THOMAS \\ S2 AI \& ML \\ Roll No.: 42}

\author{}

\date{14 FEBRUARY 2025}

\maketitle

\begin{figure}[h!]

\centering

\includegraphics[width=0.5\textwidth]{images.jpg}

\label{fig:my_label}

\end{figure}

\section{Introduction}

\subsection{What is LaTeX?}

LaTeX is a document preparation system for producing professional-looking documents. It is not a word processor. The short and simple answer is that LaTeX is a typesetting program and is an extension of the original program \texttt{TeX} written by Donald Knuth.

\subsection{What is a Typesetting Program?}

To answer this, let us look at the various stages in the preparation of a document using computers:

\begin{enumerate}

\item The text is entered into the computer.

\item The input text is formatted into lines, paragraphs, and pages.

\item The output text is displayed on the computer screen.
```

\item The final output is printed.

\end{enumerate}

A typesetting program like \texttt{TeX} is concerned only with the second stage above. So, to typeset a document using \texttt{TeX}, we type the text of the document and the necessary formatting commands in a text editor and then compile it. After that, the document can be viewed using a previewer or printed using a printer driver.

\subsection{Why Learn LaTeX?}

\begin{itemize}

\item Professional-grade typography.

\item Superior math typesetting capabilities.

\item Consistent formatting throughout a document.

\end{itemize}

\section{Comparison of Latex document with MS Word document }

% Add content for comparison here

% Table added to the same page

\begin{table}[h!]

\centering

\begin{tabular}{|p{4cm}|p{5cm}|p{5cm}|}

\hline

\textbf{Feature} & \textbf{Microsoft Word} & \textbf{LaTeX} \\ \hline

Ease of use & user friendly, WYSIWYG interface & requires knowledge of commands and syntax \\ \hline

Customization & limited to built-in formatting tools & highly customizable through coding \\

\hline

Mathematical Typesetting & basic support for equations (equation editor) & excellent support for complex equations \\ \hline

\end{tabular}

\caption{Comparison}

\end{table}

\end{document}

My First LaTeX Document

JEWEL THOMAS

S2 AI & ML

Roll No.: 42

14 FEBRUARY 2025



1 Introduction

1.1 What is LaTeX?

LaTeX is a document preparation system for producing professional-looking documents. It is not a word processor. The short and simple answer is that LaTeX is a typesetting program and is an extension of the original program TeX written by Donald Knuth.

1.2 What is a Typesetting Program?

To answer this, let us look at the various stages in the preparation of a document using computers:

1. The text is entered into the computer.
2. The input text is formatted into lines, paragraphs, and pages.
3. The output text is displayed on the computer screen.
4. The final output is printed.

A typesetting program like TeX is concerned only with the second stage above. So, to typeset a document using TeX, we type the text of the document and the necessary formatting commands in a text editor and then compile it.

After that, the document can be viewed using a previewer or printed using a printer driver.

1.3 Why Learn LaTeX?

- Professional-grade typography.
- Superior math typesetting capabilities.
- Consistent formatting throughout a document.

2 Comparison of Latex document with MS Word document

Feature	Microsoft Word	LaTeX
Ease of use	user friendly, WYSIWYG interface	requires knowledge of commands and syntax
Customization	limited to built-in formatting tools	highly customizable through coding
Mathematical Typesetting	basic support for equations (equation editor)	excellent support for complex equations

Table 1: Comparison

```

\documentclass{article}

\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{fancyhdr}
\usepackage{geometry}

\geometry{a4paper, margin=1in}

\pagestyle{fancy}

\fancyhf{}

\fancyhead[L]{Dept. of Computer Science \& Engg.}

\fancyhead[R]{Idea Lab}

\fancyfoot[C]{Page \thepage}

\begin{document}

\section*{Equations in LaTeX}

```

Mathematical equations can be written in different ways. For example, an inline equation such as $E = mc^2$ can be included within a sentence.

A standalone equation is written as follows:

```

\begin{equation}

F = ma

\end{equation}

```

Some other examples:

1. Quadratic formula:

```

\begin{equation}

```


$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

\end{equation}

2. Pythagorean theorem:

\begin{equation}

$$a^2 + b^2 = c^2$$

\end{equation}

3. A summation formula:

\begin{equation}

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

\end{equation}

\section*{Multiline Equations}

If an equation is too long, we can split it across multiple lines:

\begin{align}

$$a^2 + b^2 = c^2 \\\$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

\end{align}

\section*{Special Symbols and Characters}

\LaTeX\ allows the use of special symbols and characters such as:

\begin{itemize}

\item \textbf{Greek letters:} \(\ \alpha, \beta, \gamma, \pi, \sigma \)

\item \textbf{Set symbols:} \(\ \in, \notin, \subseteq, \supseteq \)

\item \textbf{Logical symbols:} \(\ \forall, \exists, \neg, \wedge, \vee \)

\item \textbf{Operators:} \(\ \sum, \prod, \int, \cap, \cup \)

\item \textbf{Special characters:} \# \\$ \% \& \{ \} \textasciitilde \textbackslash

\end{itemize}

\begin{flushright}

February 2025

\end{flushright}

\end{document}

Equations in L^AT_EX

Mathematical equations can be written in different ways. For example, an inline equation such as $E = mc^2$ can be included within a sentence.

A standalone equation is written as follows:

$$F = ma \tag{1}$$

Some other examples:

1. Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{2}$$

2. Pythagorean theorem:

$$a^2 + b^2 = c^2 \tag{3}$$

3. A summation formula:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \tag{4}$$

Multiline Equations

If an equation is too long, we can split it across multiple lines:

$$a^2 + b^2 = c^2 \tag{5}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{6}$$

Special Symbols and Characters

L^AT_EX allows the use of special symbols and characters such as:

- **Greek letters:** $\alpha, \beta, \gamma, \pi, \sigma$
- **Set symbols:** $\in, \notin, \subseteq, \supseteq$
- **Logical symbols:** $\forall, \exists, \neg, \wedge, \vee$
- **Operators:** $\sum, \prod, f, \cap, \cup$
- **Special characters:** $\# \$ \% \& \{ \} \backslash$

February 2025

Result- Familiarised the implementation and documentation in LaTeX

EXPERIMENT -2

Aim – Poster using Canva

Theory – Canva makes designing beautiful creative assets easy for everyone. From social media images, videos and gifs to posters, websites, booklets, multimedia presentations, and heaps more, Canva is an invaluable tool for helping you level up your design skills, even if you're just getting started.

Procedure –

1. **Start by creating a new Canva account.** You can do this by going to www.canva.com and clicking on the “Create a New Account” button. Enter your email address and choose a password. Then click on the “Create Account” button.
2. **Once you have created your account, log in.** You will be taken to the Canva dashboard. From here, you can choose to create a design from scratch, or use a template. For this tutorial, we will be creating a design from scratch.
3. **To start creating your document, click on the “Create a Design” button.** This will take you to the design editor.
4. **Once you have chosen your canvas size, you can start adding elements to your design.** To do this, simply click on the element that you want to add, and then drag it onto your canvas. You can add text, images, shapes, and lines.
5. **Once you have added all of the elements that you want, you can start styling them.** To do this, simply click on the element that you want to style, and then use the options in the toolbar that appears. You can change the font, color, size, and alignment of your element.
6. **When you are happy with your design, click on the “Download” button.** This will allow you to download your design as a PDF or JPG file. You can also share your design directly to social media or via email.

OUTPUT



RESULT- Poster using canva is successfully implemented .

EXPERIMENT – 3

Aim- Presentation using Canva

Theory- Canva makes designing beautiful creative assets easy for everyone. From social media images, videos and gifs to posters, websites, booklets, multimedia presentations, and heaps more, Canva is an invaluable tool for helping you level up your design skills, even if you're just getting started.

Procedure-

1. **Start by creating a new Canva account.** You can do this by going to www.canva.com and clicking on the “Create a New Account” button. Enter your email address and choose a password. Then click on the “Create Account” button.
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6. **When you are happy with your design, click on the “Download” button.** This will allow you to download your design as a PDF or JPG file. You can also share your design directly to social media or via email.

VIRTUAL REALITY

Let's start!



*Presented by Jewel
Thomas
S2 DIM2*

AGENDA

- | | | | |
|---|------------------------------|---|------------------------------|
| 1 | Briefly describe the concept | 5 | Briefly describe the concept |
| 2 | Briefly describe the concept | 6 | Briefly describe the concept |
| 3 | Briefly describe the concept | 7 | Briefly describe the concept |
| 4 | Briefly describe the concept | 8 | Briefly describe the concept |



WELCOME!

Virtual reality (VR) is a simulated experience that employs 3D near-eye displays and pose tracking to give the user an immersive feel of a virtual world. Applications of virtual reality include entertainment (particularly video games), education (such as medical, safety or military training) and business (such as virtual meetings). VR is one of the key technologies in the reality- virtuality continuum.

Let's begin!



FUTURE TECHNOLOGY

Our vision



The future of virtual reality is increasingly leaning towards multisensory experiences. It's not just about what users can see; it's about what they can touch, smell, and even taste. The more realistic the virtual world, the more immersive and captivating the experience for the user.



AUGMENTED REALITY

Augmented reality is an interactive experience that enhances the real world with computer-generated perceptual information. Using software, apps, and hardware such as AR glasses, augmented reality overlays digital content onto real-life environments and objects.

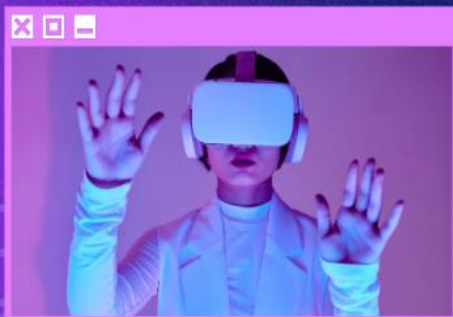
77%



Computer Modeling



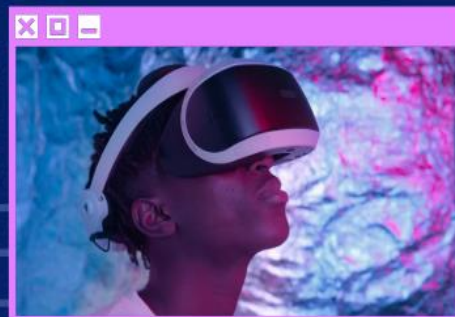
Process of constructing and manipulating computer-based mathematical, graphical or algorithmic representations of real life systems or phenomena.



Interactive Device



The device where the user can receive position, localization, navigation instructions, etc., and interact with the information.





Result- PPT using Canva is implemented successfully

EXPERIMENT-4

Aim – Design a webpage using HTML and CSS

Theory- HTML, an acronym for **HyperText Markup Language**, is the standard language for designing web pages. It is a combination of Hypertext, which establishes the links between web pages, and Markup language, which is used to define the structure of web pages within tags. HTML forms the backbone of any webpage, dictating its structure and content.

CSS (Cascading Style Sheets) is a stylesheet language used to design the webpage to make it attractive. The reason for using CSS is to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

Procedure-

<!doctype html>	Document's first declaration
<html lang="en">	Another declaration that specifies the next HTML page will be written in English.
<head>	Indicates the beginning of the heading section
<meta charset="utf-8">	Specifies the character set that is used to create the document
<title>Hello, world!</title>	The page title will appear in the browser's title bar.
<body>	The text that is planned to show on the page should be included in this area,
Hello, world!	The page's primary header
My first web page.	Regular text in paragraph
</html>	HTML document's concluding tag

few more basics about CSS properties:

CSS Properties	Description
background	Configures an element's background.

background-color	sets the element's background colour.
border	Defines an element's border.
border-color	Establishes the border's colour.
border-left	Creates the element's left boundary
border-right	Sets the element's right boundary.
border-top	Establishes the element's top boundary.
border-bottom	Defines an element's bottom boundary.
color	Sets the text colour for an element.
font	Specifies the font family, variation, weight, height, and size for a given element.
font-family	Defines the element's font family.
font-size	Specifies a font's size for an element.
font-style	Specifies a regular, italic, or oblique font style.
font-weight	Adjusts the font's weight.
height	Specifies an element's height.
margin	Determines an element's margin (outside spacing).
padding	Determines the space between an element's border and its content.
text-align	sets the text within an element's alignment.
text-shadow	Creates a text-shadow effect.
width	Determines an element's width.

<!-- ... -->	HTML comments are between the brackets. It doesn't appear on the latest page.
<meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">	Bootstrap's declaration tag. It determines webpage viewport size.
<link href="css/creative.min.css" rel="stylesheet">	It loads the Creative template's CSS stylesheet and Bootstrap's default.

Implementation –

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Library - Book List</title>
<style>
body {
font-family: Arial, sans-serif;
padding: 20px;
background-color: #f9f9f9;
}
h1 {
text-align: center;
color: #333;
}
.book-list {
display: flex;
flex-wrap: wrap;
gap: 20px;
justify-content: center;
margin-top: 40px;
}
.book {
background-color: white;
border: 1px solid #ddd;
border-radius: 10px;
padding: 15px;
width: 200px;
text-align: center;
box-shadow: 0 2px 8px rgba(0,0,0,0.1);
transition: transform 0.2s;
}
.book:hover {
transform: translateY(-5px);
}
.book img {
width: 100%;
height: auto;
border-radius: 5px;
}
.book a {
display: block;
margin-top: 10px;
text-decoration: none;
color: #007bff;
font-weight: bold;
}
.book a:hover {
text-decoration: underline;
}
</style>
</head>
<body>
```

<h1>Library Book List</h1>

<div class="book-list">

<!-- Book 1 -->

<div class="book">

The Great Gatsby

</div>

<!-- Book 2 -->

<div class="book">

1984

</div>

<div class="book">

Pride and Prejudice

</div>

</div>

</body>

</html>

BOOK1

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Pride and Prejudice - Book Details</title>
<style>
body {
font-family: Arial, sans-serif;
padding: 40px;
}
.container {
max-width: 600px;
margin: auto;
text-align: center;
}
img {
width: 300px;
border-radius: 10px;
}
h1 {
margin-top: 20px;
}
p {
line-height: 1.6;
margin-top: 10px;
}
a {
display: inline-block;
margin-top: 20px;
text-decoration: none;
background-color: #007bff;
color: white;
padding: 10px 20px;
border-radius: 5px;
}
a:hover {
background-color: #0056b3;
}
</style>
</head>
<body>
<div class="container">

<h1>Pride and Prejudice</h1>
<p><strong>Author:</strong> Jane Austen</p>
<p><strong>Published:</strong> 1813</p>
<p><strong>Summary:</strong> "Pride and Prejudice" is a romantic novel that explores the themes of love,
social standing, and individual growth. The story centers on Elizabeth Bennet, an intelligent and spirited young
woman, and her complicated relationship with the proud and wealthy Mr. Darcy. Through witty dialogue and
memorable characters, Austen critiques societal expectations and the pursuit of marriage</p>
<a href="index.html">← Back to Book List</a>
</div>
</body>
</html>
```


BOOK2

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>1984 - Book Details</title>
<style>
body {
font-family: Arial, sans-serif;
padding: 40px;
}
.container {
max-width: 600px;
margin: auto;
text-align: center;
}
img {
width: 300px;
border-radius: 10px;
}
h1 {
margin-top: 20px;
}
p {
line-height: 1.6;
margin-top: 10px;
}
a {
display: inline-block;
margin-top: 20px;
text-decoration: none;
background-color: #007bff;
color: white;
padding: 10px 20px;
border-radius: 5px;
}
a:hover {
background-color: #0056b3;
}
</style>
</head>
<body>
<div class="container">

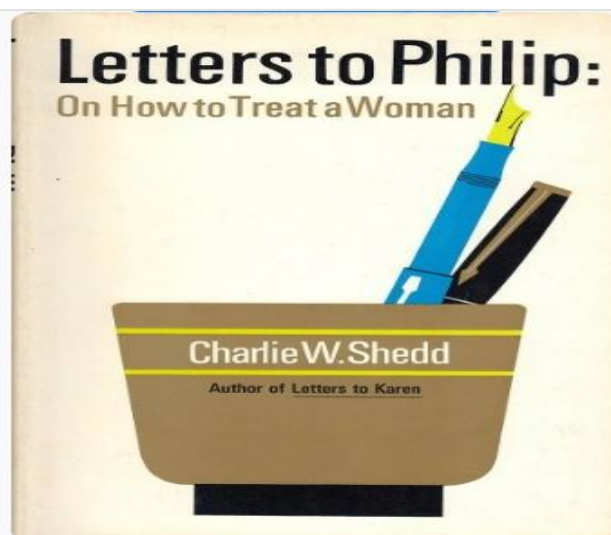
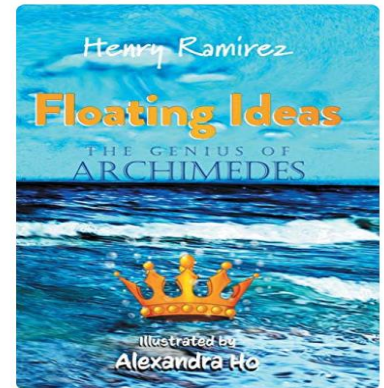
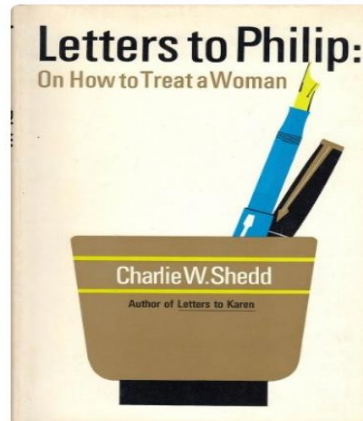
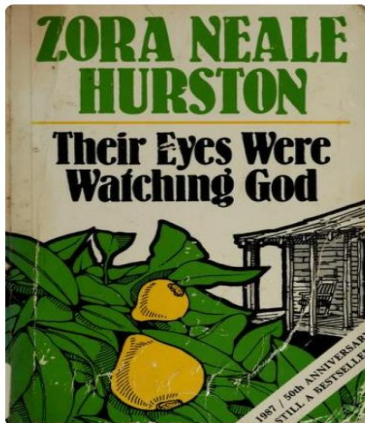
<h1>1984</h1>
<p><strong>Author:</strong> George Orwell</p>
<p><strong>Published:</strong> 1949</p>
<p><strong>Summary:</strong> 1984 is a dystopian novel set in a future totalitarian society ruled by the Party and its omnipresent leader, Big Brother. The story follows Winston Smith, a worker at the Ministry of Truth, who secretly rebels against the oppressive regime. The novel explores powerful themes like surveillance, freedom of thought, and the manipulation of truth.</p>
<a href="index.html">← Back to Book List</a>
</div>
</body>
</html>
```

BOOK3

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>The Great Gatsby - Book Details</title>
<style>
body {
font-family: Arial, sans-serif;
padding: 40px;
}
.container {
max-width: 600px;
margin: auto;
text-align: center;
}
img {
width: 300px;
border-radius: 10px;
}
h1 {
margin-top: 20px;
}
p {
line-height: 1.6;
margin-top: 10px;
}
a {
display: inline-block;
margin-top: 20px;
text-decoration: none;
background-color: #007bff;
color: white;
padding: 10px 20px;
border-radius: 5px;
}
a:hover {
background-color: #0056b3;
}
</style>
</head>
<body>
<div class="container">

<h1>The Great Gatsby</h1>
<p><strong>Author:</strong> F. Scott Fitzgerald</p>
<p><strong>Published:</strong> 1925</p>
<p><strong>Summary:</strong> A classic novel set in the Roaring Twenties, it tells the
story of mysterious millionaire Jay Gatsby and his obsession with the beautiful Daisy
Buchanan.</p>
<a href="index.html">← Back to Book List</a>
</div>
</body>
</html>
```

Library Book List



1984

Author: George Orwell

Published: 1949

Summary: 1984 is a dystopian novel set in a future totalitarian society ruled by the Party and its omnipresent leader, Big Brother. The story follows Winston Smith, a worker at the Ministry of Truth, who secretly rebels against the oppressive regime. The novel explores powerful themes like surveillance, freedom of thought, and the manipulation of truth.

[← Back to Book List](#)

Result – Designed a webpage using HTML and CSS

EXPERIMENT - 5

Aim -Familiarise Arduino microcontroller

Theory-

An Arduino microcontroller is a versatile, open-source platform used for creating interactive electronic projects. It's essentially a small, programmable computer board that allows you to interact with the physical world by reading inputs (like sensor data) and controlling outputs (like motors or LEDs). The Arduino platform includes both hardware (the board) and software (the IDE).

Core Components:

- ❑ Microcontroller: The "brain" of the Arduino, a small, programmable chip that executes instructions.
- ❑ Programming Language: Based on C and C++, but with a simplified API (Arduino language).
- ❑ Integrated Development Environment (IDE): Software used to write, compile, and upload code to the Arduino board.
- ❑ Board: The physical circuit board with various input/output pins, a microcontroller, and a USB connection for programming.

Key Features:

- ❑ Open-Source:
The hardware and software are freely available, allowing for customization and community collaboration.
- ❑ Easy to Use:
The Arduino IDE and language are designed to be accessible for beginners, even without extensive

electronics or programming knowledge.

☐ Versatile:

Arduino boards can be used for a wide range of projects, from controlling simple circuits to building complex robotic systems.

☐ Interactivity:

Arduino can interact with sensors, buttons, and other inputs, and control outputs like LEDs, motors, and other devices.

☐ Community Support:

A large and active online community provides resources, tutorials, and support for users.

Key components of an Arduino board.

☐ 1. Microcontroller - this is the brain of an Arduino, and is the component that we load

programs into. Think of it as a tiny computer, designed to execute only a specific number of things.

☐ 2. USB port - used to connect your Arduino board to a computer.

☐ 3. USB to Serial chip - the USB to Serial is an important component, as it helps translating

data that comes from e.g. a computer to the on-board microcontroller. This is what makes it

possible to program the Arduino board from your computer.

☐ 4. Digital pins - pins that use digital logic (0,1 or LOW/HIGH). Commonly used for switches

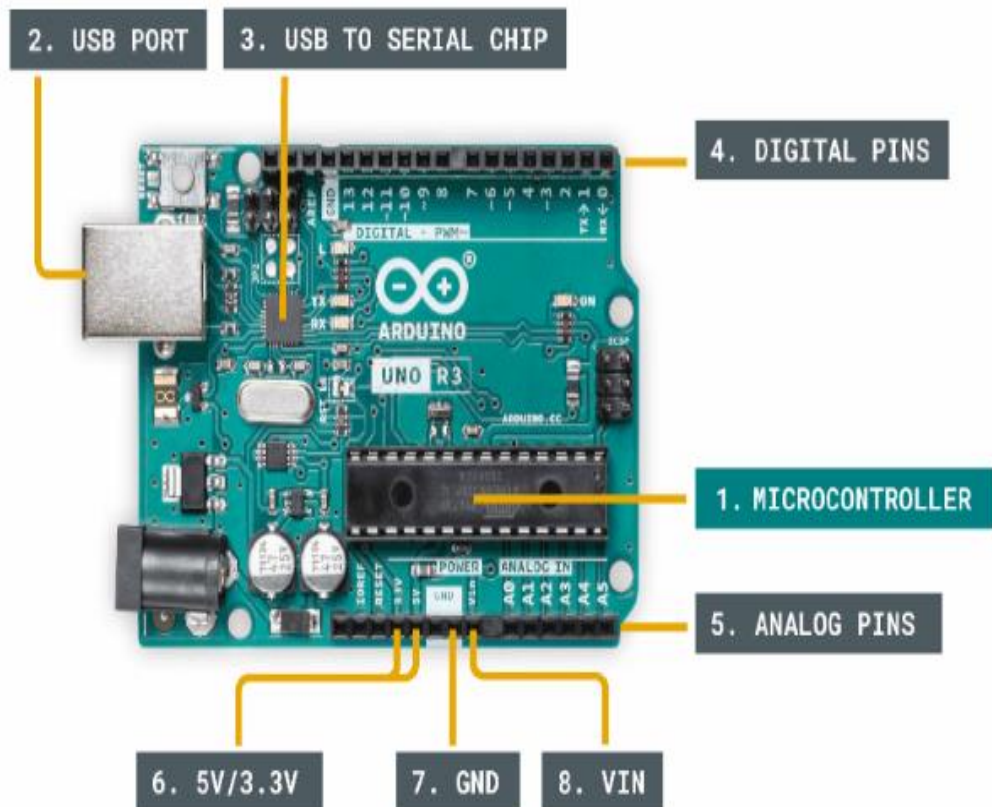
and to turn on/off an LED.

☐ 5. Analog pins - pins that can read analog values in a 10 bit resolution (0-1023).

☐ 6. 5V / 3.3V pins- these pins are used to power external components.

7. GND - also known as ground, negative or simply , is used to complete a circuit, where the electrical level is at 0 volt.

8. VIN - stands for Voltage In, where you can connect external power supplies.



Result- Familiarised Arduino Microcontroller

EXPERIMENT – 6

Aim- Blink on external LED using ARDUINO Microcontroller

Procedure-

setup() and loop() are two fundamental Arduino functions for controlling the behavior of your board. The Arduino framework automatically calls these functions, which form the foundation of any Arduino program.

The setup() function is only called once when the Arduino board boots up or is reset. Its goal is to set pin modes, initialize variables, and execute any other necessary setup tasks before the main loop begins. This function can be used to configure settings that should only be changed once over the board's lifespan.

The loop() function is the heart of an Arduino program. After the setup() function is executed, the loop() function starts running repeatedly until the Arduino is powered off or reset. It contains the main code that performs the desired tasks, controls the board, user input. Whatever is included in the loop() function will be executed in a continuous loop, allowing the Arduino to perform its intended functions continuously.

In the code, we have declared two integers, **LEDpin** and **delayT**.

LEDpin represents the pin number of the Arduino where LEDs need to be connected, and **delayT** is an integer variable for the delay() function. The delay() function accepts values in milliseconds.

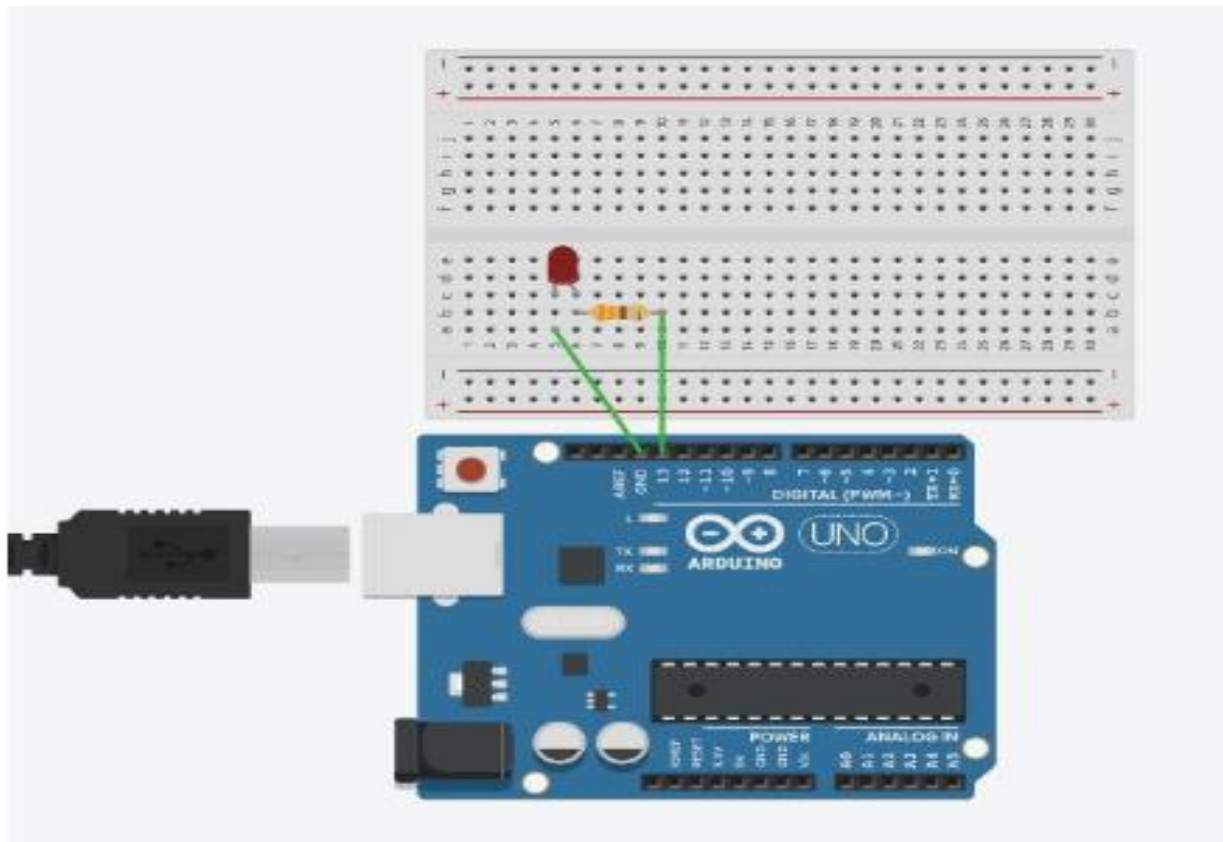
Implementation-

In the circuit diagram, we used one 330-ohm resistor in series with the [LED](#). This resistor is also called a current-limiting resistor. The Anode of the LED (the longer pin) is connected to one end of the resistor, and the cathode (the shorter pin) is connected to the ground. The other end of the resistor is connected to the Arduino pin. A step-by-step explanation is as follows:

1. **LED Connections:** Connect the LED to the breadboard. The LED has two legs, the longer of which is the anode (positive) and the shorter of which is the cathode (negative).
2. **Resistor Connection:** Insert one end of the resistor into the same row of the breadboard as the LED's Anode. The resistor's other end should be connected to the Arduino's digital output pin.
3. **Ground (GND) Connection:** Connect a jumper wire from the same row as the LED's cathode to any Arduino board GND (Ground) pin. This connects the circuit to the ground of the Arduino.

The circuit is now complete. Here's how it works:

When you upload a simple Arduino program that controls the LED, the microcontroller on the Arduino board executes the program, and the LED will blink according to the code you wrote.



Result - _Blinked an external LED using Arduino microcontroller.

EXPERIMENT – 7

Aim- Measure distance and display on serial monitor using ultrasonic sensor and Arduino microcontroller

Procedure –

Wiring:

Assuming you are using an **HC-SR04** ultrasonic sensor:

HC-SR04 Pin	Connect to Arduino Pin
VCC	5V
GND	GND
Trig	Digital Pin 9
Echo	Digital Pin 10

```
// Define pins
const int trigPin = 9;
const int echoPin = 10;

// Define variables
long duration;
float distance;

void setup() {
  // Start Serial communication
  Serial.begin(9600);

  // Set the trigPin as an Output and echoPin as an Input
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // Clear the trigPin by setting it LOW
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  // Trigger the sensor by setting the trigPin HIGH for 10 microseconds
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the echoPin, returns the sound wave travel time in microseconds
  duration = pulseIn(echoPin, HIGH);

  // Calculate the distance
  distance = duration * 0.034 / 2; // Speed of sound wave divided by 2 (go and back)

  // Print the distance on the Serial Monitor
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // Small delay before next measurement
  delay(500);
}
```

How it works:

- **Trigger** pin sends an ultrasonic burst.
- **Echo** pin listens for the bounce back.
- **pulseIn()** measures the time for the echo.
- The distance is calculated by $(\text{duration} \times \text{speed of sound}) / 2$.

Result - Measured distance and display on serial monitor using ultrasonic sensor and Arduino microcontroller